

PRPM — Probability Representation Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Probability & Uncertainty Standard (PPUS)

Operational Layer: Prediction Probability & Uncertainty Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Probability Representation

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AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Probability Representation

Canonical Definition

Probability Representation Module (PRPM) defines the governance conditions under which probability associated with a Governed Prediction is expressed in a form that preserves its predictive meaning, methodological origin, applicable horizon, boundary, conditions, and provenance.

PRPM governs how probability becomes an explicit predictive representation without allowing a number, percentage, likelihood label, score, odds value, or transformed quantity to imply a meaning that has not been methodologically established.

It transforms:



Governed Prediction + Probability Source + Applicable Probability Methodology

into:

Governed Probability Representation

Its foundational principle is:

A probability value is not self-explanatory.

Operational Role

PRPM is the first internal governance space of the Prediction Probability & Uncertainty Standard (PPUS) .

It receives a specific Governed Prediction from PFOS and establishes how probability associated with that prediction may legitimately be represented.

The focused operational question is:

What does this probability actually mean, where did it come from, and what prediction does it qualify?

The governed progression is:

Probability Source → Probability Meaning → Claim Association → Representation Form → Horizon / Boundary / Condition Association → Provenance → Semantic Integrity → Governed Probability Representation

The primary output is:

Governed Probability Representation

which becomes an input to the remaining PPUS governance process.

Internal Module Flow

Probability Representation → Predictive Uncertainty → Confidence Representation → Predictive Range Governance → Conditional Probability

Current Position: Probability Representation

Probability Representation

A Probability Representation expresses estimated likelihood associated with a specific Predictive Claim according to an applicable methodology.

It may take forms such as:

decimal probability; percentage; odds; probability category; qualitative likelihood; probability range; distributional representation; conditional probability.

The representation must remain semantically connected to the prediction it qualifies.

Probability Object

Every represented probability should have an identifiable probability object.

Example:

72%

is insufficient by itself.

A governed representation may establish:

72% estimated probability that Event X occurs within 30 days under Conditions A and B.

Probability–Claim Association

Probability must remain associated with the specific Predictive Claim it qualifies.

A probability attached to one claim must not silently migrate to another claim.

Example:

70% probability of abnormal vibration threshold exceedance

must not become:

70% probability of machine failure

unless a governed methodology establishes that relationship.

Probability Association $\neq$ Semantic Transfer

Probability Value

Where probability is numerical, the value should remain explicit.

Examples:

0.72

72%

These representations may be mathematically equivalent where the same probability scale applies.

Their display format does not change their underlying probability meaning.

Probability Scale

Probability representations must use an identifiable scale.

Common forms include:

0–1

or:

0–100%

Systems must not assume that every 0–1 value is probability.

Score $\neq$ Probability

AI and analytical systems frequently produce numerical scores between 0 and 1.

Examples include:

anomaly scores; classification scores; similarity scores; ranking scores; model confidence; normalized indicators.

Their numerical scale does not make them probabilities.

0–1 Scale $\neq$ Probability Scale by Default

Probability Source

The source of probability must remain identifiable.

Possible sources include:

statistical model; Bayesian methodology; frequency-based estimation; expert elicitation; simulation; prediction market; collective forecasting process; machine-learning model; calibrated score; hybrid methodology.

Probability Provenance

Probability Provenance records the origin through which the represented probability was established.

It should permit later determination of:

where the probability came from

and:

what methodology gives the number its meaning.

Same Probability Value ≠ Same Predictive Meaning

Two predictions may both state:

70%

while originating from fundamentally different processes.

Example:

Prediction A:

70% — calibrated statistical estimate

Prediction B:

70% — expert judgment

Prediction C:

70% — market-implied probability

Prediction D:

70% — transformed AI score

The number alone does not make these representations equivalent.

Methodological Association

A probability representation must remain traceably associated with the methodology that supports it.

Where no methodology supports interpreting a value as probability, PRPM must not allow the value to acquire probability semantics merely through labeling.

Probability Label ≠ Probability Methodology

Direct Probability Output

Some predictive methodologies directly produce probability estimates.

Example:

$P(\text{Event X}) = 0.64$

The direct origin should remain identifiable.

Derived Probability

Probability may be derived from another representation.

Examples:

odds → probability

calibrated score → probability

market price → implied probability

distribution → event probability

The derivation must remain traceable.

Derived Probability ≠ Source Value

A transformed probability must not be represented as though the source system directly produced it.

Probability Transformation

A transformation may include:

calibration; normalization; odds conversion; price conversion; score conversion; distribution integration; category mapping.

Material transformations require a governed transformation reference.

Transformation Validity

The existence of a mathematical conversion does not establish predictive legitimacy.

A transformation must preserve or legitimately establish probability meaning.

Probability Format

A probability may be displayed as:

0.73

73%

73 in 100

where equivalent meaning is preserved.

Formatting should not change semantic interpretation.

Display Probability

The displayed probability may differ in precision from the canonical probability value.

Example:

Canonical:

72.64%

Display:

73%

Both should remain traceably associated.

Display Value ≠ Canonical Probability Value

Where rounding or simplification is material, systems should preserve the underlying governed probability representation.

Probability Precision

The number of decimal places should reflect methodological support.

Example:

67.43821%

may create false precision where the methodology only supports:

approximately 65–70%.

Numerical Precision $\neq$ Predictive Precision

Probability Rounding

Rounding should not materially distort probability interpretation.

This is particularly important around:

decision thresholds; category thresholds;

escalation thresholds; automated-action thresholds.

Probability Threshold

Where probability triggers a governance or operational threshold, both the underlying value and applicable threshold should remain distinguishable.

Example:

Canonical probability:

49.6%

Display:

50%

Decision threshold:

$\geq 50\%$

The rounded display must not silently change the operational state.

Point Probability

A Point Probability expresses a single estimated value.

Example:

68%

It must not imply that uncertainty around the estimate is zero.

Point Probability $\neq$ Zero Uncertainty

Probability Range

Where probability is represented as a range:

60–75%

the meaning of that range must remain explicit.

PRGM governs range mechanics in greater depth.

PRPM ensures that the range remains identifiable as a probability representation.

Probability Distribution

Where a full distribution supports the prediction, PRPM may preserve a distribution reference rather than forcing the prediction into one point probability.

Qualitative Probability Representation

Where numerical probability is not appropriate, a prediction may use qualitative terms such as:

Unlikely

Possible

Likely

Highly Likely

Such terms require governed meaning where they materially affect interpretation.

Likelihood Label ≠ Universal Probability Range

A label such as:

Likely

may have different definitions across organizations.

The applicable mapping must therefore remain identifiable.

Probability Category Mapping

Where numeric probability is converted into categories, the mapping must be explicit.

Example:

0–20% → Very Low

>20–40% → Low

>40–60% → Moderate

>60–80% → High

>80–100% → Very High

These thresholds are implementation-specific unless otherwise governed.

Probability Horizon

Probability must remain associated with the Prediction Horizon.

Example:

70% probability of failure

is incomplete if it is unclear whether the horizon is:

24 hours; 30 days; five years.

Same Probability + Different Horizon ≠ Same Predictive Representation

Probability Boundary

Where material, probability must preserve the applicable Predictive Boundary.

Example:

70% probability of demand growth

may apply to:

one region; one customer segment; one product; the global market.

Probability Conditions

Where probability depends upon conditions, those conditions must remain associated.

Example:

75% probability of completion by December if financing is secured by June.

Removing the condition changes the predictive meaning.

Probability Scenario

Scenario-dependent probabilities must remain distinguishable.

Example:

Base Scenario: 65%

Adverse Scenario: 30%

Favorable Scenario: 85%

These must not be averaged or collapsed without separate governance.

Conditional Probability

Conditional probability expresses probability under a defined condition.

Canonical representation:

$P(X | Y)$

The condition is part of the probability meaning.

CPM governs conditional probability in depth.

PRPM ensures that conditionality is not lost in representation.

Probability Without Condition $\neq$ Conditional Probability

Removing the condition creates a different representation.

Probability and Assumption Association

Probability may depend materially upon Predictive Assumptions governed under PAS.

Where relevant, the probability representation should retain association with those assumptions.

Probability and Prediction Basis

The probability may depend upon information governed under PBAS.

PRPM preserves the relevant basis association without re-governing the Prediction Basis.

Probability and Methodology

PMES establishes Prediction Methodology.

PRPM preserves the methodology that gives probability its predictive meaning.

Probability and Prediction Formation

PFOS establishes what the prediction claims.

PRPM does not redefine that claim.

It attaches probability meaning to the already formed prediction.

Probability Representation $\neq$ Predictive Claim Formation

Probability and Confidence

A probability value must not be labeled as confidence unless the applicable methodology establishes that meaning.

CFRM separately governs confidence representation.

Probability $\neq$ Confidence

Probability and Uncertainty

A point probability does not eliminate uncertainty.

PUNM governs Predictive Uncertainty.

Probability Representation $\neq$ Complete Uncertainty Representation

Probability and Range Governance

PRGM governs ranges, intervals, bounds, precision, and related representation in greater depth.

PRPM preserves the probability semantics of any range used.

Probability and Feasibility

Probability does not establish whether an outcome is operationally feasible.

PFES governs Prediction Feasibility.

Probability $\neq$ Feasibility

Probability and Consequence

Probability does not establish the consequence of an event.

A 1% event may have catastrophic consequences.

A 90% event may have negligible consequences.

Probability $\neq$ Consequence

Probability and Risk

Probability may contribute to risk analysis.

PRPM does not govern general risk determination.

Market-Implied Probability

Prediction markets may derive implied probability from market prices.

The resulting value should remain explicitly identifiable as:

Market-Implied Probability

unless additional governance legitimately establishes another meaning.

Market-Implied Probability $\neq$ Governed Probability Estimate by Default

Market Price $\neq$ Probability

Market prices may reflect:

information; liquidity; incentives; participant composition; hedging; speculation; market rules; capital constraints.

Conversion into implied probability does not remove those characteristics.

Expert Probability

Expert judgment may legitimately produce probability.

The representation should preserve that origin.

Expert Probability $\neq$ Statistical Probability by Default

Collective Probability

Probability formed from multiple predictors must preserve its collective origin.

CPGS governs legitimacy of Collective Prediction composition.

PRPM governs representation once a probability exists.

Average Probability

A mathematical average of several probabilities is not automatically a legitimate collective probability.

Average Probability $\neq$ Governed Collective Probability by Default

Machine-Generated Probability

A machine may generate probability automatically.

Machine generation does not itself establish governance sufficiency.

Classifier Confidence

Classification systems may produce confidence values.

These should remain distinguishable from real-world event probability.

Classification Confidence $\neq$ Predictive Probability

Anomaly Score

An anomaly score may indicate deviation from learned patterns.

It does not automatically represent:

probability of future failure.

Probability Semantic Integrity

Semantic Integrity exists where the displayed probability preserves:

what event/state it qualifies; source; methodology; horizon; boundary; conditions; probability type.

Probability Semantic Distortion

Distortion occurs where representation changes predictive meaning.

Examples:

Model Score $\rightarrow$ Event Probability

Conditional Probability $\rightarrow$ Unconditional Probability

Market-Implied Probability $\rightarrow$ Objective Probability

Expert Confidence $\rightarrow$ Statistical Probability

Probability Representation Conflict

A conflict exists where multiple incompatible probability representations are associated with the same Predictive Claim without sufficient distinction.

Example:

Model: 72%

Expert: 41%

The disagreement does not need to be eliminated.

Its origins must remain visible.

Probability Disagreement ≠ Governance Failure

Different legitimate methodologies may produce different probability estimates.

PRPM preserves the disagreement rather than manufacturing artificial consensus.

Probability Representation State

A representation may occupy states such as:

Established

Conditionally Established

Limited

Conflicting

Insufficiently Supported

Unable to Determine

Unknown Probability

Where no legitimate probability can be established:

Probability Unable to Determine

is a valid governed result.

Unknown Probability ≠ 50%

A probability of 50% is an actual probability estimate.

Ignorance is not automatically equivalent to equal likelihood.

No Numerical Probability Required

Not every prediction requires a numerical probability.

Where methodology supports only qualitative representation, PRPM should not manufacture numerical precision.

Numerical Representation ≠ Better Governance by Default

Historical Probability Representation

The probability associated with the prediction at a particular lifecycle state must remain historically reconstructable.

Later probability changes must not silently overwrite earlier values.

Current Probability ≠ Historical Probability

Probability Representation Version

Material changes to probability representation should remain distinguishable through versioning or equivalent historical mechanisms.

Representation Correction

A display or metadata error may be corrected without necessarily changing the underlying probability estimate.

Example:

Canonical probability:

0.72

incorrect display:

7.2%

corrected display:

72%

Representation Correction $\neq$ Probability Revision by Default

Probability Representation Governance Requirements

PRPM requires that:

1. every probability representation remain associated with a specific Governed Prediction; 2. the applicable Predictive Claim remain identifiable; 3. probability meaning remain explicit; 4. probability source remain identifiable; 5. probability provenance remain traceable; 6. methodology supporting probability remain identifiable; 7. probability value or qualitative state remain explicit; 8. applicable probability scale remain identifiable; 9. scores remain distinguishable from probabilities; 10. classifier confidence remain distinguishable from event probability; 11. derived probabilities remain distinguishable from source values; 12. material transformations remain traceable; 13. canonical and displayed values remain distinguishable where material; 14. false precision be avoided; 15. material rounding effects remain governable; 16. point probability not imply zero uncertainty; 17. qualitative likelihood terms have governed meaning where material; 18. category mappings remain identifiable; 19. Prediction Horizon remain associated; 20. Predictive Boundary remain associated where material; 21. conditions remain preserved; 22. scenario-dependent probabilities remain distinguishable; 23. probability remain distinguishable from confidence; 24. probability remain distinguishable from feasibility;

25. probability remain distinguishable from consequence; 26. market-implied probability remain properly identified; 27. expert probability preserve its origin; 28. collective probability preserve its collective provenance; 29. conflicting probability estimates remain distinguishable; 30. inability to determine probability remain a legitimate state; 31. unknown probability not default to 50%; 32. historical probability representations remain reconstructable.

Governance Logic

PRPM applies the following governance logic:

Governed Prediction

Probability Source

Applicable Probability Methodology

↓

Probability Object Identified

↓

Predictive Claim Association Established

↓

Probability Source / Provenance Identified

↓

Probability Type Identified

↓

Value / Qualitative Representation Established

↓

Scale / Format Defined

↓

Prediction Horizon / Boundary Associated

↓

Conditions / Scenario Associated

↓

Transformation / Derivation Assessed

↓

Score / Confidence / Probability Distinction Verified

↓

Precision / Rounding Assessed

↓

Semantic Integrity Assessed

↓

Conflicts / Representation Gaps Identified

↓

Probability Representation State Established

↓

Historical Representation Preserved

↓

Governed Probability Representation

↓

PUNM — Predictive Uncertainty Module

Where sufficient representation cannot be established:

Probability Representation Gap → Clarification / Qualification / Reduced Precision / Conditional Representation / Probability Unable to Determine

Minimum Implementation Framework

A minimum implementation of PRPM must establish:

A. Prediction Reference

The applicable Governed Prediction.

B. Predictive Claim Reference

The specific claim qualified by the probability.

C. Probability Identifier

A distinguishable probability representation reference.

D. Probability Source

At minimum:

Statistical / Bayesian / Expert / Simulation / Market-Implied / Machine-Generated / Collective / Hybrid / Other

E. Probability Type

At minimum:

Point / Range / Distributional / Qualitative / Conditional / Scenario-Dependent / Other

F. Probability Value / Representation

The actual numerical or qualitative representation.

G. Probability Scale

Where applicable.

H. Methodology Reference

The methodology supporting the probability meaning.

I. Prediction Horizon

The applicable temporal horizon.

J. Predictive Boundary

Where material.

K. Condition / Scenario Reference

Where applicable.

L. Transformation / Derivation Reference

Where the probability was derived from another value.

M. Canonical / Display Representation

Where the displayed representation differs materially from the canonical value.

N. Representation State

At minimum:

Established / Conditionally Established / Limited / Conflicting / Insufficiently Supported / Unable to Determine

O. Representation Version / Historical State

The applicable version or historical representation reference.

Governance Outputs

PRPM may produce:

Governed Probability Representation; Probability Representation Record; Probability Identifier; Probability Value; Probability Type; Probability Scale; Probability Source Record; Probability Provenance Record; Methodology Association; Claim Association; Horizon Association; Boundary Association; Condition

Association; Scenario Association; Probability Transformation Record; Probability Derivation Record; Canonical Probability Value; Display Probability Value; Probability Precision Record; Probability Rounding Record; Probability Category Mapping; Probability Semantic Integrity State; Probability Representation Conflict; Probability Representation Gap; Probability Representation State; Probability Representation Version; historical probability representation.

The primary governed output is:

Governed Probability Representation

which becomes an input to:

PUNM — Predictive Uncertainty Module.

Critical Distinctions

A Probability Value Is Not Self-Explanatory

Probability Association $\neq$ Semantic Transfer

Score $\neq$ Probability

0–1 Scale $\neq$ Probability Scale by Default

Same Probability Value $\neq$ Same Predictive Meaning

Probability Label $\neq$ Probability Methodology

Derived Probability $\neq$ Source Value

Display Value $\neq$ Canonical Probability Value

Numerical Precision $\neq$ Predictive Precision

Point Probability $\neq$ Zero Uncertainty

Likelihood Label $\neq$ Universal Probability Range

Same Probability + Different Horizon $\neq$ Same Predictive Representation

Probability Without Condition $\neq$ Conditional Probability

Probability Representation $\neq$ Predictive Claim Formation

Probability $\neq$ Confidence

Probability Representation $\neq$ Complete Uncertainty Representation

Probability $\neq$ Feasibility

Probability $\neq$ Consequence

Market-Implied Probability $\neq$ Governed Probability Estimate by Default

Market Price $\neq$ Probability

Expert Probability $\neq$ Statistical Probability by Default

Average Probability $\neq$ Governed Collective Probability by Default

Classification Confidence $\neq$ Predictive Probability

Probability Disagreement $\neq$ Governance Failure

Unknown Probability $\neq$ 50%

Numerical Representation $\neq$ Better Governance by Default

Current Probability $\neq$ Historical Probability

Enterprise Application

PRPM provides a common probability language for organizations operating predictive systems.

It is especially relevant where probability moves between:

AI models; analysts; dashboards; APIs; forecasting platforms; management systems; automated decision systems.

Without Probability Representation Governance, a number can change meaning as it moves between systems.

A:

0.82 model score

can become:

82% probability

and later:

82% certainty

without any methodology ever establishing those transitions.

PRPM preserves what the probability actually means.

Use Case 1 — A 0.87 AI Score Becomes an “87% Failure Probability”

An industrial AI system produces:

Anomaly Score: 0.87

The score represents the degree to which current sensor patterns differ from the model's learned normal operating profile.

A dashboard developer assumes that 0.87 means:

87% probability of equipment failure within 30 days

and displays that statement to maintenance teams.

No calibrated transformation between anomaly score and failure probability exists.

The critical governance question is:

Can a numerical value on a 0–1 scale legitimately become an event probability merely because it looks like one?

PRPM requires the original value to remain identified as an anomaly score unless a governed methodology establishes a legitimate probability transformation.

0–1 Scale ≠ Probability Scale by Default

Use Case 2 — The Same 70% Comes From Three Different Sources

A strategic planning team receives three predictions that a new product will reach its first-year sales target.

Statistical model: 70%

Executive expert estimate: 70%

Prediction market implied probability: 70%

The dashboard presents all three simply as:

Probability: 70%

Management interprets the identical numbers as three independent confirmations of the same probability.

The critical governance question is:

Can identical probability values be treated as equivalent when their predictive origins are materially different?

PRPM requires each representation to preserve its source, methodology, provenance, horizon, conditions, and probability type.

The values may legitimately be compared.

Their identical numerical representation does not make their predictive meaning identical.

Same Probability Value $\neq$ Same Predictive Meaning

Canonical Boundary

PRPM governs the identification, meaning, source, provenance, methodological association, value, type, scale, format, transformation, derivation, precision, rounding, horizon, boundary, conditionality, scenario association, semantic integrity, state, versioning, and historical preservation of probability representations associated with a Governed Prediction. It does not govern Predictive Uncertainty itself, Confidence Representation itself, detailed Predictive Range Governance, Conditional Probability mechanics, Prediction Feasibility, Collective Prediction composition, communication and reliance, prediction-level revision, outcome determination, or predictive performance assessment.

PRPM establishes:

what the represented probability means.

It does not establish:

whether the predicted outcome will occur.

Canonical Closing Statement

A probability becomes governable only when its number, meaning, source, methodology, conditions, and predictive object remain connected.

PRPM prevents scores, percentages, market signals, confidence values, and transformed quantities from silently acquiring probability meanings they were never designed to carry.

The same number can represent different predictions; governance begins by preserving the difference.

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